

Assignment 2 Solutions

Modern Cryptography (2)

Problem 1. For the Schnorr signature scheme, reusing the same random nonce r to sign two distinct messages is *not secure*. Let the two signatures be

$$\sigma_1 = (R, z_1), \quad \sigma_2 = (R, z_2),$$

where the same commitment $R = g^r$ is used in both signatures. For Schnorr signatures,

$$z_i = r + e_i s \pmod{p},$$

where s is the signing key and $e_i = H(R, M_i)$ is the challenge for message M_i . Hence

$$z_1 - z_2 = (e_1 - e_2)s \pmod{p}.$$

If $e_1 \neq e_2$, then the secret key can be recovered as

$$s = (z_1 - z_2)(e_1 - e_2)^{-1} \pmod{p}.$$

Since $M_1 \neq M_2$, we expect $e_1 \neq e_2$ except with negligible probability (assuming the hash function behaves like a random oracle or is collision resistant in the relevant sense). Therefore, reusing the same nonce immediately reveals the secret key, and the scheme becomes completely insecure.

Problem 2. Consider the variant of DSA in which the hash function H is omitted, so the response is computed as

$$z = r^{-1}(M + ds) \pmod{p},$$

where $d = F(R)$ and $R = g^r$. We show that this scheme is existentially forgeable.

An adversary can forge a valid signature on a message of its choice as follows. Choose arbitrary values

$$a, b \in \mathbb{Z}_p, \quad b \neq 0,$$

and define

$$R := g^a h^b.$$

Then query the random oracle F at R and obtain

$$d := F(R).$$

Now set

$$z := db^{-1} \pmod{p}, \quad M := az \pmod{p}.$$

We claim that (d, z) is a valid signature on M . Indeed,

$$(g^M h^d)^{z^{-1}} = (g^{az} h^d)^{z^{-1}} = g^a h^{dz^{-1}}.$$

Because $z = db^{-1}$, we have $dz^{-1} = b$, and therefore

$$(g^M h^d)^{z^{-1}} = g^a h^b = R.$$

Also, by construction, $d = F(R)$. Hence the verification equation is satisfied. So the adversary outputs the forged signature

$$\sigma = (d, z)$$

on message M , without knowing the secret key s .

Therefore, omitting H makes the scheme insecure.

Problem 3. We show that if the Discrete Logarithm problem is hard in G , then the DSA identification scheme is UI-PA secure.

Assume toward contradiction that there exists a PPT adversary $\mathcal{A}$ that breaks UI-PA security with non-negligible advantage

$$\text{Adv}_{\text{UI-PA}}^{\mathcal{A}}(\lambda) = \Pr[\mathcal{A} \text{ successfully impersonates}] = \varepsilon(\lambda),$$

where $\varepsilon(\lambda)$ is non-negligible. We construct a PPT algorithm $\mathcal{B}$ that solves the discrete logarithm problem.

Algorithm $\mathcal{B}$ is given (G, p, g, h) with $h = g^s$, and its goal is to recover s . It runs $\mathcal{A}$ and simulates all passive transcripts requested by $\mathcal{A}$. An accepting transcript of the identification protocol has the form

$$(R, e, d, z)$$

with

$$R = (g^e h^d)^{z^{-1}}.$$

Hence $\mathcal{B}$ can generate a perfectly distributed accepting transcript by choosing

$$e, d \xleftarrow{\$} \mathbb{Z}_p, \quad z \xleftarrow{\$} \mathbb{Z}_p^*, \quad R := (g^e h^d)^{z^{-1}}.$$

This has exactly the same distribution as a real protocol transcript, so the simulation in the passive phase is perfect.

After seeing polynomially many such transcripts, $\mathcal{A}$ outputs a first message R^* . Now $\mathcal{B}$ sends a random challenge $(e, d_1) \xleftarrow{\$} \mathbb{Z}_p \times \mathbb{Z}_p$. If $\mathcal{A}$ answers with z_1 such that

$$R^* = (g^e h^{d_1})^{z_1^{-1}},$$

then $\mathcal{B}$ rewinds $\mathcal{A}$ to the point immediately after R^* was sent, keeps all internal randomness of $\mathcal{A}$ fixed, and sends a fresh independent challenge $(e, d_2) \xleftarrow{\$} \mathbb{Z}_p \times \mathbb{Z}_p$ with the same e and a new d_2 . If $\mathcal{A}$ also answers with z_2 such that

$$R^* = (g^e h^{d_2})^{z_2^{-1}},$$

then $\mathcal{B}$ extracts s .

Indeed, write $R^* = g^r$. From the two accepting transcripts we obtain

$$r \equiv (e + d_1 s) z_1^{-1} \equiv (e + d_2 s) z_2^{-1} \pmod{p}.$$

Multiplying through gives

$$(e + d_1 s) z_2 = (e + d_2 s) z_1,$$

and therefore

$$e(z_2 - z_1) = (d_2 z_1 - d_1 z_2) s \pmod{p}.$$

So whenever $d_2 z_1 - d_1 z_2 \neq 0$, we can compute

$$s = e(z_2 - z_1)(d_2 z_1 - d_1 z_2)^{-1} \pmod{p}.$$

If $d_2 z_1 - d_1 z_2 = 0$, then the above equality implies $e(z_2 - z_1) = 0$. For $e \neq 0$, this forces $z_1 = z_2$, and then from

$$e + d_1 s \equiv e + d_2 s \pmod{p}$$

we get $(d_1 - d_2)s \equiv 0$. Since $s \neq 0$ except with probability $1/p$, this would imply $d_1 = d_2$. Thus, except with probability at most $2/p$, the denominator is nonzero and extraction succeeds.

It remains to analyze the success probability. Let ω denote all randomness used by $\mathcal{A}$ and by $\mathcal{B}$ before the final challenge is chosen. For fixed ω , define

$$V(\omega, e, d) = 1$$

if and only if, when the final challenge is (e, d) , the adversary outputs a valid response z for the already fixed first message R^* . Then

$$\varepsilon = \Pr_{\omega, e, d}[V(\omega, e, d) = 1].$$

Now consider the rewinding experiment in which we sample the same ω and the same e , but two independent values $d_1, d_2 \xleftarrow{\$} \mathbb{Z}_p$. Let $\varepsilon_{\mathcal{B}}$ be the probability that both rewind runs accept and $d_1 \neq d_2$. Then

$$\varepsilon_{\mathcal{B}} \geq \Pr_{\omega, e, d_1, d_2}[V(\omega, e, d_1) = 1 \wedge V(\omega, e, d_2) = 1] - \Pr[d_1 = d_2].$$

Since $\Pr[d_1 = d_2] = 1/p$, we get

$$\varepsilon_{\mathcal{B}} \geq \sum_{\omega} \Pr[\omega] \left(\frac{1}{p} \sum_{e \in \mathbb{Z}_p} \Pr_d[V(\omega, e, d) = 1] \right)^2 - \frac{1}{p}$$

by independence of d_1, d_2 . Using Jensen's inequality (equivalently Cauchy-Schwarz),

$$\sum_{\omega} \Pr[\omega] \left(\frac{1}{p} \sum_e \Pr_d[V(\omega, e, d) = 1] \right)^2 \geq \left(\sum_{\omega} \Pr[\omega] \cdot \frac{1}{p} \sum_e \Pr_d[V(\omega, e, d) = 1] \right)^2 = \varepsilon^2.$$

Hence

$$\varepsilon_{\mathcal{B}} \geq \varepsilon^2 - \frac{1}{p}.$$

Since p is exponential in the security parameter, $1/p$ is negligible; therefore if ε is non-negligible, then $\varepsilon_{\mathcal{B}}$ is also non-negligible. Combining this with the previous extraction step, $\mathcal{B}$ solves the discrete logarithm problem with non-negligible probability, contradicting the DL assumption.

Therefore, the DSA identification scheme is UI-PA secure.

Problem 4. We prove that if the identification scheme is UI-PA secure and H, F are modeled as random oracles, then the Fiat-Shamir signature scheme is EUF-CMA secure.

1. The setup. Let

$$\text{PK} = (G, p, g, h = g^s)$$

be the public key, where $s \in \mathbb{Z}_p$ is the secret key.

We consider two algorithms:

- **Forger A:** a PPT adversary that attempts to output a valid forgery $(M^\dagger, d^\dagger, z^\dagger)$, where $M^\dagger$ was not previously submitted to the signing oracle.

- **Simulator B :** a PPT algorithm that runs A as a subroutine and tries to break the UI-PA security of the identification protocol.

Algorithm B receives PK , but does not know the secret key s .

Suppose that A makes at most q_H queries to H and at most q_F relevant queries / table entries for F . As before, B chooses two indices uniformly at random:

$$i_H \in \{1, \dots, q_H\}, \quad i_F \in \{1, \dots, q_F\}.$$

Intuitively, B guesses that the final forgery of A will involve the i_H -th fresh query to H and the i_F -th relevant fresh entry in the table T_F . If this guess is correct, then B will be able to embed the final UI-PA challenge into exactly the pair $(M^\dagger, R^\dagger)$ associated with the forgery.

2. Simulating the oracles for A . The simulator B must answer A 's queries to the random oracles H, F , as well as signing queries. It maintains two tables:

$$T_H \subseteq \mathcal{M} \times \mathbb{Z}_p, \quad T_F \subseteq G \times \mathbb{Z}_p.$$

These tables ensure consistency of oracle answers.

3. Simulation of the random oracles.

H-oracle queries. When A queries $H(M)$, algorithm B proceeds as follows:

- If M already appears in T_H , return the previously recorded value.
- Otherwise, if this is not the distinguished i_H -th fresh query to H , choose

$$e \xleftarrow{\$} \mathbb{Z}_p,$$

record $(M, e) \in T_H$, and return e .

- If this is the distinguished i_H -th fresh query, then B will associate this message with the final challenge value e^* .

F-oracle queries. When A queries $F(R)$, algorithm B proceeds as follows:

- If R already appears in T_F , return the previously recorded value.
- Otherwise, if this is not the distinguished i_F -th relevant fresh entry for F , choose

$$d \xleftarrow{\$} \mathbb{Z}_p,$$

record $(R, d) \in T_F$, and return d .

- If this is the distinguished i_F -th relevant fresh entry, then B sends this value $R^* := R$ to its UI-PA challenger as the first message of the impersonation attempt. The challenger replies with a random challenge pair

$$(e^*, d^*) \xleftarrow{\$} \mathbb{Z}_p^2.$$

B then records

$$(R^*, d^*) \in T_F$$

and sets the distinguished H -query to the value e^* .

Thus the two programmed values are consistent with the same final challenge:

$$H(M^\star) = e^\star, \quad F(R^\star) = d^\star.$$

4. Simulation of signing queries. When A asks for a signature on a message M , the simulator B constructs a valid signature directly by programming the random oracle F .

More precisely, B proceeds as follows:

1. Obtain

$$e := H(M)$$

from the simulated H -oracle.

2. Choose

$$d \xleftarrow{\$} \mathbb{Z}_p, \quad z \xleftarrow{\$} \mathbb{Z}_p^\star.$$

3. Compute

$$R' := (g^e h^d)^{z^{-1}}.$$

4. If $R' \notin T_F$, set

$$T_F[R'] := d.$$

If $R' \in T_F$ already with a different value, then B aborts. This is a bad event of negligible probability.

5. Return the signature

$$(d, z)$$

to A .

This signature is valid, because verification recomputes

$$R' = (g^e h^d)^{z^{-1}}$$

and then checks

$$F(R') = d,$$

which holds by construction.

5. Using the forgery. Suppose that A finally outputs a valid forgery

$$(M^\dagger, d^\dagger, z^\dagger),$$

where $M^\dagger$ was never submitted to the signing oracle. Let

$$e^\dagger := H(M^\dagger)$$

and define

$$R^\dagger := (g^{e^\dagger} h^{d^\dagger})^{(z^\dagger)^{-1}}.$$

Since the forgery is valid, we have

$$F(R^\dagger) = d^\dagger.$$

The important point is that, except with negligible probability, the value $R^\dagger$ must already have an entry in the table T_F . Indeed, if $R^\dagger$ had never appeared in T_F , then $F(R^\dagger)$ would still be an independent uniform random value in $\mathbb{Z}_p$, and the event

$$F(R^\dagger) = d^\dagger$$

would occur with probability at most $1/p$, which is negligible.

Hence, except with negligible probability, the forgery corresponds to some value $R^\dagger$ whose F -value has already been fixed in the simulation, either because A explicitly queried $F(R^\dagger)$, or because B previously programmed $F(R^\dagger)$ while answering a signing query.

Now assume the good event that:

1. $M^\dagger$ is the distinguished i_H -th fresh query to H ,
2. $R^\dagger$ is the distinguished i_F -th relevant fresh entry in T_F ,
3. no abort occurs during the simulation.

Then by construction,

$$e^\dagger = e^*, \quad d^\dagger = d^*, \quad R^\dagger = R^*.$$

Therefore,

$$R^* = (g^{e^*} h^{d^*})^{(z^\dagger)^{-1}},$$

so B can answer the UI-PA challenger with

$$z^* := z^\dagger$$

and wins the impersonation game.

6. Probability analysis. If A succeeds with probability $\varepsilon(\lambda)$, then, except for negligible bad events, the final forgery determines a corresponding pair $(M^\dagger, R^\dagger)$ with

$$R^\dagger \in T_F.$$

Algorithm B succeeds whenever it guesses correctly both the distinguished H -query and the distinguished relevant F -entry associated with the forgery. Since these two guesses are uniform and independent, this happens with probability at least

$$\frac{1}{q_H} \cdot \frac{1}{q_F} = \frac{1}{q_H q_F}.$$

Therefore,

$$\Pr[B \text{ wins UI-PA}] \geq \frac{\varepsilon(\lambda)}{q_H q_F} - \text{negl}(\lambda).$$

Since q_H and q_F are polynomially bounded, a non-negligible forgery probability $\varepsilon(\lambda)$ would imply a non-negligible success probability for B , contradicting the UI-PA security of the identification scheme.

Therefore, under the UI-PA security of the identification protocol, the Fiat–Shamir signature scheme is EUF-CMA secure in the random oracle model.